

Securin Security Analyst Interview Preparation Roadmap

Module 1: Computer Networks (Highest Priority)

Topic 1.1: OSI Model

Subtopics

- 7 Layers
- Layer Functions
- Encapsulation
- Decapsulation
- Protocol Mapping

Important Questions

1. Explain the OSI Model.
2. What happens at each layer?
3. Which layer does ARP belong to?
4. Which layer does TCP belong to?
5. Which layer does HTTP belong to?
6. Difference between Layer 2 and Layer 3.
7. What is encapsulation?
8. What is decapsulation?

Topic 1.2: TCP/IP

Subtopics

- TCP
- UDP
- Reliability
- Flow Control
- Sequence Numbers

Important Questions

1. Difference between TCP and UDP.
2. Explain TCP Three-Way Handshake.
3. What are SYN, SYN-ACK and ACK?
4. Why is TCP reliable?
5. Why is UDP faster?
6. What is retransmission?
7. What is packet loss?
8. What is flow control?

Topic 1.3: DNS

Subtopics

- DNS Resolution
- DNS Hierarchy
- Recursive Resolver
- Records

Important Questions

1. How DNS works?
2. What happens when you enter google.com?
3. What is Port 53?
4. What is an A Record?
5. What is a CNAME Record?
6. What happens if DNS fails?
7. Difference between DNS and DHCP.

Topic 1.4: HTTP & HTTPS

Subtopics

- HTTP Request
- HTTP Response
- Methods
- Headers
- Status Codes

Important Questions

1. Difference between HTTP and HTTPS.
2. Explain GET.
3. Explain POST.
4. Explain PUT.
5. Explain PATCH.
6. Explain DELETE.
7. Difference between PUT and PATCH.
8. Difference between 401 and 403.
9. Difference between 404 and 500.
10. What is TLS?

Topic 1.5: Networking Devices

Subtopics

- Router

- Switch
- Firewall
- Proxy
- VPN
- NAT

Important Questions

1. Router vs Switch.
2. What is a Firewall?
3. What is NAT?
4. What is VPN?
5. Why do organizations use VPNs?
6. What is a Proxy Server?

Module 2: APIs & Web Technologies

Topic 2.1: API Fundamentals

Subtopics

- APIs
- REST
- Endpoints
- Resources
- Stateless Communication

Important Questions

1. What is an API?
2. What is REST API?
3. Difference between SOAP and REST.
4. What is an Endpoint?
5. Why are REST APIs stateless?
6. What is a Resource?

Topic 2.2: JSON

Subtopics

- Objects
- Arrays
- Nested Structures

Important Questions

1. What is JSON?

2. Difference between Object and Array.
3. How do APIs exchange data?
4. Parse a nested JSON response.

Topic 2.3: Authentication & Authorization

Subtopics

- Basic Auth
- API Keys
- JWT
- OAuth

Important Questions

1. Authentication vs Authorization.
2. What is JWT?
3. Explain JWT structure.
4. What is OAuth?
5. Why use Bearer Tokens?
6. Session vs JWT.

Topic 2.4: API Testing

Subtopics

- Postman
- Curl
- Request Validation
- Response Validation

Important Questions

1. How do you test APIs?
2. API returns 401. What will you check?
3. API returns 403. What will you check?
4. API returns 404. What will you check?
5. API returns 500. What will you check?
6. API timeout troubleshooting steps.
7. API works in Postman but not in code. Why?

Module 3: Cybersecurity Fundamentals

Topic 3.1: Security Principles

Subtopics

- CIA Triad
- Defense in Depth
- Least Privilege
- Attack Surface

Important Questions

1. Explain CIA Triad.
2. What is Defense in Depth?
3. What is Least Privilege?
4. What is an Attack Surface?
5. Why is security layered?

Topic 3.2: Risk Concepts

Subtopics

- Threat
- Vulnerability
- Risk
- Exploit

Important Questions

1. Threat vs Vulnerability.
2. Risk vs Vulnerability.
3. What is an Exploit?
4. Give examples of vulnerabilities.
5. How do organizations manage risk?

Module 4: Web Application Security

Topic 4.1: SQL Injection

Subtopics

- SQLi
- Prepared Statements
- Input Validation

Important Questions

1. What is SQL Injection?
2. How SQL Injection works?
3. How to prevent SQL Injection?
4. What are Prepared Statements?

Topic 4.2: XSS

Subtopics

- Stored XSS
- Reflected XSS
- DOM XSS

Important Questions

1. What is XSS?
2. Types of XSS.
3. How to prevent XSS?
4. Why is output encoding important?

Topic 4.3: CSRF

Subtopics

- CSRF Tokens
- Session Abuse

Important Questions

1. What is CSRF?
2. How does CSRF work?
3. How do CSRF Tokens work?
4. How to prevent CSRF?

Module 5: Network Security

Topic 5.1: Common Attacks

Subtopics

- MITM
- Packet Sniffing
- DDoS
- Phishing

Important Questions

1. What is MITM?
2. What is Packet Sniffing?
3. What is DDoS?
4. Difference between DoS and DDoS.
5. How is DDoS mitigated?
6. What is Phishing?

Module 6: Cryptography

Topic 6.1: Encryption

Subtopics

- Symmetric Encryption
- Asymmetric Encryption
- Public Key
- Private Key

Important Questions

1. Symmetric vs Asymmetric Encryption.
2. What is AES?
3. What is RSA?
4. Public Key vs Private Key.
5. Why is RSA slower?

Topic 6.2: SSL/TLS

Subtopics

- Certificates
- TLS Handshake
- HTTPS

Important Questions

1. What is SSL/TLS?
2. Explain TLS Handshake.
3. How HTTPS works?
4. Why are certificates needed?
5. What is a CA (Certificate Authority)?

Module 7: Vulnerability Management

Topic 7.1: Vulnerability Assessment

Subtopics

- CVE
- CVSS
- Severity Ratings
- Prioritization

Important Questions

1. What is CVE?

2. What is CVSS?
3. How are vulnerabilities prioritized?
4. What is Risk Scoring?
5. Which vulnerability would you fix first and why?

Module 8: Programming & Project Defense

Topic 8.1: Programming Fundamentals

Subtopics

- Strings
- Arrays
- Collections
- OOP
- JSON Parsing

Important Questions

1. Reverse a String.
2. Count Character Frequency.
3. Find Duplicate Elements.
4. Parse JSON Response.
5. Read and Process a Log File.

Topic 8.2: Project Discussion

Subtopics

- Architecture
- APIs
- Database
- Security Design

Important Questions

1. Explain your project.
2. Explain your architecture.
3. Why did you choose this design?
4. Which APIs are used?
5. How is authentication implemented?
6. How is authorization implemented?
7. How would you secure your application?
8. What challenges did you face?

Final Mock Interview Question

Every student must answer this without notes:

Explain what happens when a user opens <https://www.google.com> in a browser.

A strong answer should naturally cover:

- DNS
- IP Resolution
- TCP Handshake
- TLS Handshake
- HTTP Request
- HTTP Response
- Certificates
- Encryption
- Browser Rendering